Outline

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1. Design Requirements

This section describes all requirements that the hardware team must meet to consider a successful delivery of the hardware prototype.

1.1.Design Functionality

The hardware prototype shall offer the following features:

1). Location tracking

The device shall use GNSS such as GPS, BeiDou, etc, depending on operation area to provide real-time geolocation data

2). Body motion tracking

The device shall use an IMU sensor to provide user motion data: 3-axis acceleration and 3-axis gyroscope.

3). Biometric data tracking

The device shall use biometric sensors to track the following data:

- Heart rate (HR)
- blood oxygen saturation (SpO2)
- pulse rate
- body temperature
- heart-rate variability (HRV)
- respiration rate (RR)
- advanced analytics such as sleep quality and stress level information

4). Dual-way phone call

The device shall support dual-way phone calls via 4G LTE using telecom module(s).

5). Emergency contact

The device shall offer a button for users to press to initiate an emergency call.

6). Battery status

The device shall provide battery status using a LED which lights in 3 colors:

- Green: full
- Yellow: half-level
- Red: low

7). Device Pairup

The device shall provide device pair-up using BLE.

1.2 Performance Requirements

The device shall be power efficient so that the system is powered only once per week. The battery charging circuit shall not take a large area.

1.3 Physical Requirements

The prototype must not exceed 3.5 cm x 3.5 cm x 1.3 cm. And the prototype shall be of light weight so that users can wear the prototype by touching it on chest in a similar way as Biobutton:

https://www.youtube.com/watch?v=hj-KsfzRpN0

The prototype can look like the following:



2. Component list

The device shall contain at least the following components:

- nRF5340
- MAX32674 algorithm and sensor hub

- MAX86176
- MAX30208 body temperature sensor
- BMI270
- SIM7500
- Battery
- Power management IC
- Antenna design
- LED
- Button
- Audio speaker
- Audio codec
- eSim

Please note that if any component needs to be replaced, there must be a strong reasons, typically for the following:

- Smaller area
- Better performance
- Lower power

3. General Requirements

1).Weekly update meeting for project sync-ups

2).Summary report for each milestone deliverable.